Behavior-Driven Planning of Electric Truck Charging Infrastructure for Intercity Operations

Qiujun Qian, Mi Gan, Lifei Wei, Zhu Yao, and Yuxin He

***Abstract*— Truck stop behavior plays a critical role in supporting powertrain electrification, which is key to reducing emissions in road freight. However, it is still unclear whether these stops meet the charging needs for electric trucks during intercity trips. This study aims to address this issue by analyzing intercity truck travel behavior based on trajectory data to inform battery electric truck (BET) technology design and charging infrastructure planning. The study proposes a data-driven truck trip identification framework based on GPS data to obtain truck stops and truck trip ends. Empirical analysis based on gasoline-enabled truck drivers' travel behaviors in China shows that a 400 km battery range will meet almost 90% of the continuous driving needs of intercity truck trips. It is recommended that en-route charging stations be designed with a charging power of more than 564 kW, in line with the parking time characteristics of 80% of truck stops. The study also identifies potential power supply shortages on the intercity road power grid between 11:00-13:00, 18:00-19:00, and 21:00-1:00. By quantifying the number of facilities needed to provide BET drivers an energy-viable path to complete intercity trips without excessive detour costs, the study found that the travel behavior-based optimization model can reduce the level of truck charging detour to 5% by selecting just a small subset of alternative charging facility locations (55/302). Our study demonstrates that analyzing intercity truck travel behavior can provide critical insights for designing BET technology and charging infrastructure to support the transition to low-emission road freight.**

***Index Terms*— Truck stop behavior, truck electrification for intercity trips, truck trip identification, charging facility layout.**

I. Introduction

ROAD freight has always been an essential mode of transportation in many countries and regions. In 2019, road freight in China carried more than 70% of the total freight volume [1], while in the US, road freight accounted for 72% of the volume [2]. As a result of globalization and economic development, it is foreseeable that the demand for road freight will continue to grow. However, fuel trucks contribute significantly to environmental pollution and toxic emissions, the transport sector in the US accounts for 28% of greenhouse gases, nearly 56% of NOX [3]. In the context of global "dual carbon" initiative, powertrain electrification of trucks is a critical measure to reduce emissions, eliminate exhaust pollutants, and help achieve public health and the greenhouse gas reduction targets [4].

In recent years, various efforts have been made in academia and practical fields to facilitate the electrification of trucks. However, the adoption of electric heavy-duty trucks for intercity trips faces several challenges. The primary challenge is the battery technology, which currently has limitations in terms of energy density and weight, resulting in higher battery costs, reduced payload capacity and limited driving range for electric heavy-duty trucks. Additionally, the limited availability and distribution of charging infrastructure are significant barriers to the adoption of electric heavy-duty trucks, as they require longer charging times due to their larger battery capacity. The limited driving range of electric heavy-duty trucks may also negatively impact the feasibility of freight operations. Furthermore, electric heavy-duty trucks require larger, heavier batteries and more complex electric drivetrain systems than conventional diesel trucks, resulting in higher manufacturing costs and possibly higher prices for customers.

Many scholars and enterprises have studied different methods to tackle these problems, such as optimizing the design of the powertrain, improving battery capacity and charging efficiency, and developing wireless charging technology. However, the main bottleneck of current electric vehicles is the size of the battery and its high price, resulting in battery-range limitations. Insufficient battery range and insufficient charging infrastructure are real issues that need to be addressed to accelerate the development of truck electrification. To achieve this goal, it is necessary to consider not only technological advancements but also the synchronicity between the charging needs of electric trucks and the inherent travel and stopping requirements of truck drivers during intercity trips. Understanding these factors can aid in the development of more effective technical strategies. However, current research mainly focuses on whether the electric power from a single charge can meet the entire intercity trip, and there is a lack of attention on the feasibility of electric truck charging and battery swapping

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during multiple stops throughout the journey. By considering the power demand between two successive charging stops instead of the entire trip, the minimum power requirement for electric trucks can be greatly reduced, which is highly beneficial for the electrification of intercity freight transportation. Therefore, understanding the travel and stopping needs of electric truck drivers is crucial for the successful adoption of intercity electric trucks and the realization of a sustainable freight transportation system.

To address the knowledge gap regarding the travel requirements of electric truck drivers for intercity transportation, this paper proposes a novel planning framework for battery electric truck (BET) technology design and charging infrastructure planning based on mining truck stop behavior with application of on-road GPS data. The framework aims to identify truck travel behavior and gather information on stops using GPS data, enabling an analysis of the compatibility between truck travel behavior and the requirements of electric vehicle technology. Specifically, the study explores the battery-range needs, charging power needs, and the number of charging facilities required for intercity trips based on the distribution of travel distance, dwell time, and trip distribution. This information helps to identify the technical requirements of electric trucks from the perspective of truck drivers' travel needs, which can assist relevant decision-makers in formulating effective electric truck technology strategies and infrastructure construction strategies.

The rest of the paper is organized as follows. Section II reviews related literature, while section III presents the flowchart, database, trip identification method, and optimal location model for truck charging infrastructure. Section IV describes the validation results of the proposed truck trip identification method and the empirical analysis of requirements related to battery range limitations, charging power of charging facilities, and the layout of charging facilities for intercity truck trips. Finally, section V concludes the paper.

II. Related Works

A. Identify Truck Trips Based on GPS Data

The current methods for truck trip identification primarily involve two main steps: identifying truck stops and determining the truck trip ends based on these stops. Previous studies have utilized a speed threshold approach to identify truck stop events. For instance, some studies have considered trajectory points with zero velocity as truck stops [5], [6], [7], while others have taken into account speeds below a specific threshold [8], [9], [10], [11]. Additionally, certain studies have incorporated truck engine data to infer truck stops, identifying trajectory points where the engine is turned off as truck stops [12], [13], [14]. Another approach utilized distance thresholds to identify dwelling points as truck stops [15].

The second step involves identifying the end of truck trips. Truck loading and unloading processes typically require a significant amount of time [16], while the duration of stays en-route is relatively short, often due to factors such as congestion and refueling [17]. As a result, many studies identify trip ends using dwell time thresholds. Some studies set the threshold at 5 minutes [8], 3 minutes [5], [18], taking into consideration factors such as traffic signal cycles, refueling time, and road traffic conditions. While some studies utilize longer thresholds, such as 2 hours [6], 30 min [11], [19], 26 min [7], and 10 min [9], based on empirical knowledge. Additionally, a few studies employ supervised learning methods [10], [20], or calculate the entropy value of truck stops using information about the truck carrier identifier [15] to infer truck trip ends.

B. Analysis of Technical Requirements of Electric Trucks

Due to the long-distance and heavy load transportation characteristics of trucks, technical requirements of electric trucks have become a growing interest within the research community. Previous studies have investigated various aspects including the impact of electric trucks on grid load, the charging rate requirements for these vehicles, and other related factors. For example, in [4], the impacts of additional electrical loads on distribution systems were explored when heavy-duty electric vehicle charging is implemented. In [21], the charging speed requirements for semi-trailer electric trucks have been simulated. They found that the megawatt-level charging speed accounts for a larger share of the energy demand for long-haul trucks. In [22], by using of high-power fast-charging for battery-electric trucks, they found that smaller batteries can also provide feasibility for long-distance transportation scenarios, and the economy per ton-kilometer of battery-electric trucks increases with greater weight.

C. Deployment of Charging Stations

The location and layout of charging stations have a significant impact on the endurance of electric trucks, the investment cost of operators, and the user satisfaction of electric drivers, which has garnered attention from researchers. Several studies aim to maximize the coverage of charging infrastructure. For instance, in [23], the authors proposed an optimization approach for charging station locations for short-haul trucks based on a maximum coverage model in South East Queensland. In a similar vein, in [24], a two-step optimization method was proposed for deploying charging stations, combining spatial statistics and maximum coverage location model. In [25], the design and operation of charging stations have been explored by integrating agent-based charging station models with vehicle schedules derived from real vehicle telemetry data analysis.

Some studies focus on minimizing detour behavior for electric vehicle drivers, which is more practical, particularly in highway networks with poor road connectivity. For example, in [26], the authors proposed the optimal location problem for intercity truck charging stations that aims to minimize the driver detour mileage in highway networks. In [27], based on the background of long-distance intercity truck travel, a two-layer hybrid nonlinear integer programming charging station location model was developed that minimizes driver detours. In [28], the authors constructed a charging station location model that minimizes charging distance, and optimizes the layout of facilities within the station using queuing theory.

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Additionally, driver waiting time for charging is a crucial indicator of satisfaction. In [29], the authors used a centralized charging station database (CSDB) to coordinate charging between vehicles, minimizing driver waiting time at charging stations. In [30], the authors established a dual-objective model with the objective functions including minimizing the total cost and reducing charging time.

D. Research Gaps and Contributions

The literature review above indicates some gaps that need to be addressed in order to provide more realistic insights on electric truck penetration. (1) the study on inferring the origin-destination (OD) of truck travel using GPS data fails to consider the important role of vehicle heading change in OD identification. (2) detailed truck behavior data is crucial for understanding the operational patterns of conventional trucks and inferring the charging behaviors and requirements for BET. However, there is a limited understanding of BET adoption behavior in the truck sector due to a lack of available large-scale truck operating data. (3) The existing research on the deployment of charging stations relies on theoretical models and lacks the application of real truck travel data, which is of great significance for enhancing the realism of the research.

To address the aforementioned research gaps, this paper proposes a novel planning framework for BET technology design and charging infrastructure planning based on mining truck stop behavior with application of on-road GPS data. The contributions of this paper are as follows:

(1) We enhance the intercity truck trip identification method based on GPS data, taking into account vehicle heading changes. Furthermore, the effectiveness of this method is validated using real waybill data, and the results show that the method adopted in this paper brings a better effect.

(2) This paper first attempts to empirically analyze the technical needs of BET including battery range requirement, charging power requirement, and charging facilities location from the perspective of truck travel behavior.

(3) We conduct experiments to determine the facility layout that provides BET drivers with an energy-efficient route for intercity trips, minimizing detour costs. Our findings reveal that the travel behavior-based charging facilities layout design could reduce truck charging detours to just 5%, by selecting a small subset of alternative charging facility locations (55 out of 302).

III. Methodology

A. Flowchart

In this study, we first conducted data pre-processing and then utilized truck GPS data to identify truck stops and trip ends, enabling us to gather information about truck trips. Secondly, we analyzed the synchronicity between truck travel behavior and charging requirements to inform the technology design of BET. Finally, based on truck travel characteristics, we developed a model for determining optimal locations for en-route charging stations for intercity truck trips. The flowchart is shown in Fig. 1.

TABLE I
Trajectory Dataset Introduction

Item	Example	Description
User-id	1_2488617	The unique identification code of the truck
Timestamp	2018/1/1 0:09:05	The format is yyy-mm-dd-hh:mm:ss
Latitude	100.255206	The format is ddd.dddd, the unit is degrees
longitude	25.595998	The format is ddd.dddd, the unit is degrees

B. Dataset and Pre-Processing

With the emergence of China's online freight exchange (OFEX) Platform, cargo owners can publish shipping needs through the online platform anytime, anywhere, while individual truck drivers can filter and view interested cargo sources, and get in touch with cargo owners through the platform. The OFEX Platform has accumulated a large amount of the driver's mobile GPS data of trucks, as well as trip information of online transactions through mobile APPs. We obtained the above two types of data from China's largest road logistics Internet information platform.

1) Trajectory Dataset: We obtained 1.1 million GPS trajectories of 978 heavy trucks over the period from Dec.16, 2017 to Feb.8, 2018 in southwest China from the OFEX platform. The trajectory data field includes the truck driver ID, timestamp, and GPS coordinates (longitude and latitude), as shown in Table I. Fig.2 displays the spatial distribution of the trajectory dataset, with a focus on the four provincial administrative regions of Yunnan, Guizhou, Sichuan, and Chongqing in China. By dividing the area into a 4km $*$4km grid, the raster color indicates the density of GPS recording points, the redder the color of the raster, the more GPS recording points it contains. It can be seen that the hot spots of truck trajectories are primarily in Chengdu, Chongqing, Kunming, and Guiyang, which are all provincial capitals/municipalities with relatively prosperous economic development [31].

Due to various factors such as geographical environment, GPS equipment, and signals, trajectory data often encounters problems such as data duplication, data drift, and data loss. Therefore, it is necessary to perform data cleaning on the trajectory dataset. We employ the following two main steps for data cleaning:

(1) Duplicate records are identified as GPS points that have consistent time and are located within a distance of less than 300 meters from each other. In such cases, only the first duplicate record is retained [32].

(2) GPS points with a speed of more than 200km/h [6] were screened, which accounts for 5.13% of the total dataset. Since these abnormal speed GPS points may be caused by weak signals, we apply the mean smoothing method to correct them. This involves replacing the outlier with the mean coordinate and timestamp of the four GPS points before and after it.

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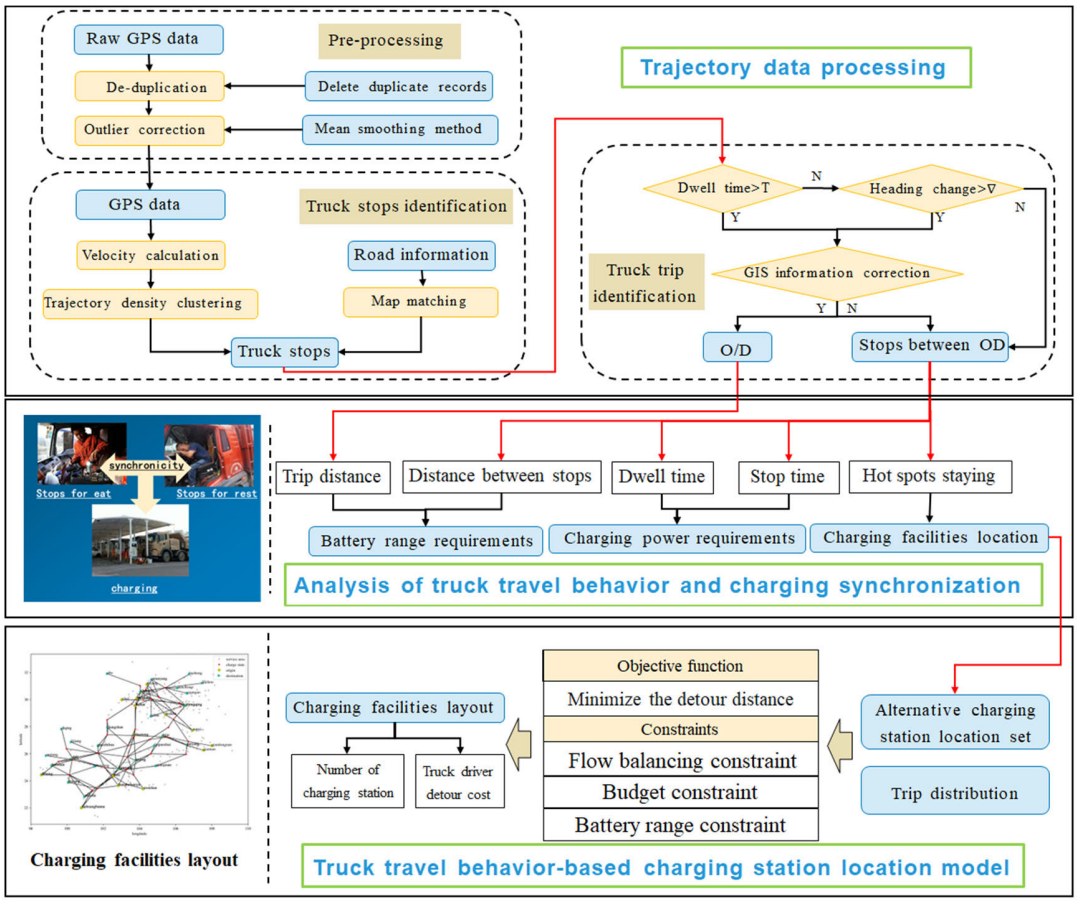


Fig. 1. Flowchart of methodology.

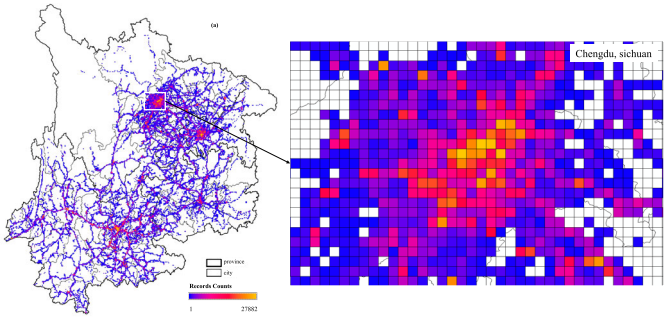


Fig. 2. Geographic density distribution of the truck trajectory.

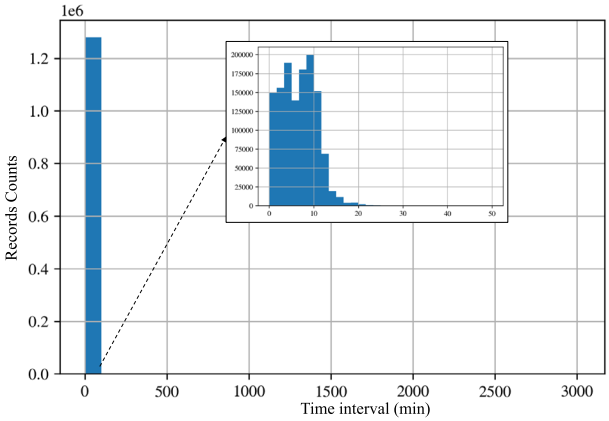


Fig. 3. Trajectory point interval time distribution.

After cleaning the trajectory data, we examine the time interval between adjacent GPS points, as shown in Fig.3. It can be observed that the majority of elapsed times between successive GPS pings are below 10 minutes, with only 2.02% of the data having intervals greater than 15 minutes.

2) Trip Dataset: In addition to the trajectory data, we have also obtained trip datasets, which are often challenging to acquire in practical scenarios [33]. This trip dataset contains 17248 trip records of 813 truck drivers from Jan.1 to Jun.10, 2018 in southwestern China. The trip records data field includes the Online transaction id, User-id, Origin, Destination, Departure time, and Arrival time, as presented in Table II. Similar to the trajectory data, the geographic coverage of the trip dataset encompasses four provincial administrative regions: Yunnan, Guizhou, Sichuan, and Chongqing. Fig.4 visualizes the freight interaction between two cities, the redder the color of the line the more trips. To ensure consistency, we extract a subset of the trip dataset that aligns with the used-id and recording time in the trajectory dataset. After eliminating irrelevant data, the final trip dataset used in this paper comprises 961 trip records from 213 truck drivers, spanning from January 1st to February 8th, 2018. The objective is to

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TABLE II
TRIP DATASET INTRODUCTION

Item	Example	Description
Online transaction id	769a41*****5 2844	The unique identification code of the online transaction
User-id	1_2488617	The unique identification code of the truck
Origin	Qujing, Yunnan	The departure city of the waybill
Destination	Kunming, Yunnan	The arrival of the waybill to the city
Departure time	2018/1/4 17:04:08	The timestamp when the truck first arrived at the origin
Arrival time	2018/1/5 19:44:36	The timestamp when the truck first arrived at the destination

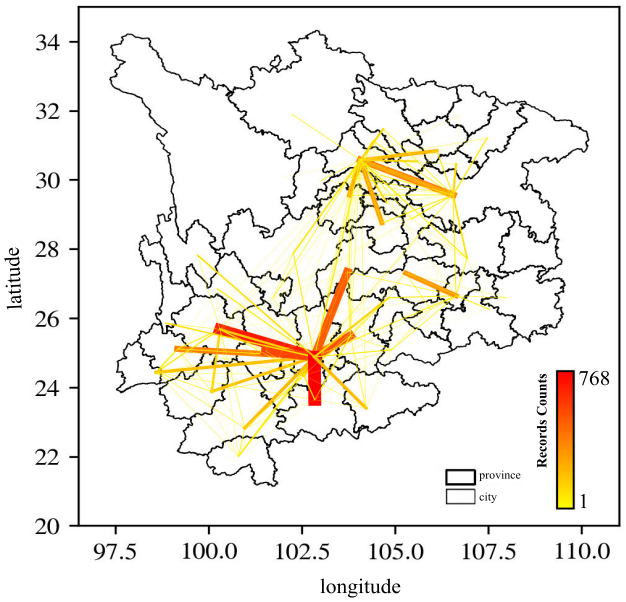


Fig. 4. The geographic distribution of the trip dataset.

calibrate the identified trip ends with their corresponding trip records in this paper.

C. Truck Trip Identification Algorithm

There are two primary challenges in trip identification. The first challenge involves identifying truck stops. We begin by extracting segments of heavy truck idle GPS points from the entire GPS trajectory using a predetermined speed threshold. Subsequently, we utilize the Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm to identify truck stops.

The second challenge is selecting the appropriate stop as the end of the trip. In order to achieve this, we employ two criteria. Firstly, we utilize dwell time to identify the end of the truck trip. Additionally, we analyze the heading change of

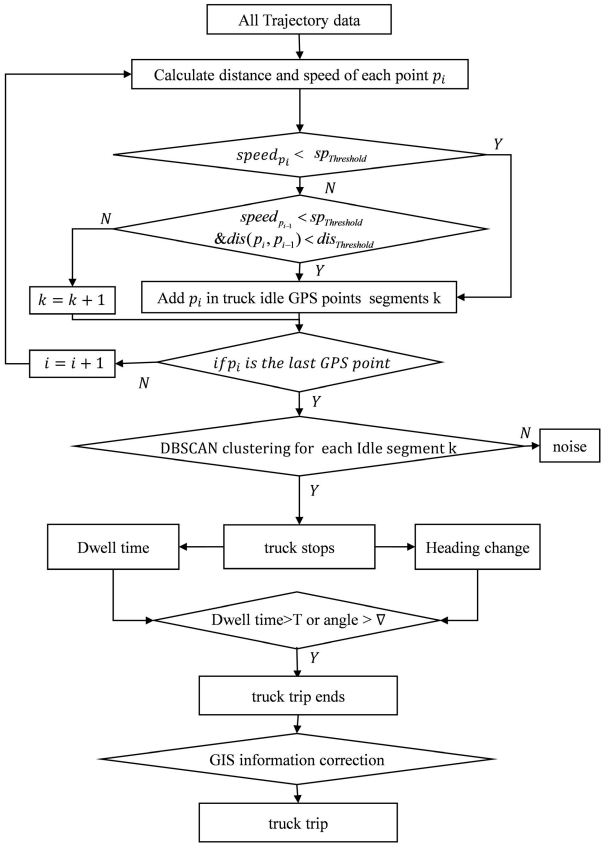


Fig. 5. Truck trip identification algorithm.

truck stops and consider stops with a larger heading change as the end of the trip. Furthermore, to account for the possibility of extended stop times during the journey (e.g., overnight stays in service areas), we optimize the end of the truck trip using geographic information. The truck trip identification algorithm is illustrated in Fig. 5.

1) Truck Stops Identification: The initial step involves the identification of truck stops. Consistent with prior research, heavy truck stops are identified by applying a predetermined speed and interval threshold to their entire GPS trajectory. This study employs the Great-Circle algorithm to calculate the geographical distance between successive GPS pings. By utilizing this distance and the elapsed time, the speed of heavy trucks is determined. It should be noted that the driver's activity event typically occurs when the vehicle is not in a high-speed driving state. In accordance with China's Road Traffic Safety Law, the speed limit for non-motorized vehicles is set at 15km/h. Consequently, a speed threshold $sp_{Threshold}$ of 15km/h is adopted. This threshold is comparatively higher than the thresholds used in previous studies, which include 5km/h [8], 8km/h [11], [19], and 10km/h [9]. The rationale behind selecting a larger speed threshold is that it reduces the impact of speed error. In the subsequent step, trajectory density recognition will be employed to accurately identify the stop points. Therefore, the use of a larger speed threshold is justified.

At the same time, the issue of trajectory point drift needs to be considered, as it can lead to abnormal speed readings.

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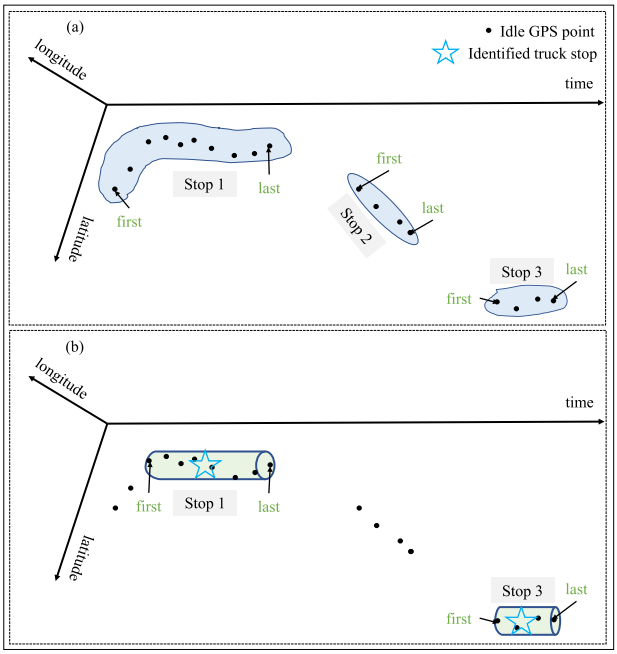


Fig. 6. A case study of truck stop identification.

Therefore, if a GPS point is in a high-speed point but has only moved a limited distance within the distance threshold, it is still regarded as an idle point. Keeping in line with the aforementioned speed threshold, a truck moving at 15 km/h would theoretically take just over 15 minutes to cover a distance of 3.75 km, a distance threshold $dis_{Threshold}$ of 3.75 km was used. By traversing all the GPS points, the idle trajectory segment of the truck can be obtained. It should be noted that these threshold values are not fixed and can be adjusted based on the characteristics of the truck's GPS samples.

When trucks engage in activities such as loading, unloading, or refueling, their trajectory points tend to exhibit a more concentrated pattern in space. In Fig.6(a), we observe three idle trajectory segments labeled as stop1, stop2, and stop3. However, only a subset of the GPS points within stop1 display spatial aggregation, while stop2 does not exhibit a gathering effect in space, only all the GPS points within stop3 display spatial aggregation. This implies that not all points within stop1 are actively engaged, whereas stop2 represents a congested period. To further identify truck stops, we employ the DBSCAN algorithm, commonly used for clustering trajectories [34]. DBSCAN defines a cluster as the largest set of densely connected points, capable of partitioning trajectories into clusters with sufficient density and identifying clusters of any shape in noisy spatial datasets. There are two important parameters in the DBSCAN algorithm: Eps and MinPtS, Eps is the neighborhood radius when defining density, and MinPts is the threshold when defining core points. By applying DBSCAN specifically to the truck's idling trajectory segments, any GPS points that cannot be clustered are considered congestion stays, as illustrated in Fig.6(b).

In DBSCAN, clustering results can vary depending on different parameter settings. The Silhouette Coefficient is a

TABLE III
TRAJECTORY CLUSTER ALGORITHM PARAMETER SELECTION

Eps (km)	MinPts	Number of Clusters	Average Silhouette Weight
0.5	2	1179	0.811
1.5	2	1179	0.857
2.0	**2**	**1179**	0.884*
2.5	2	1179	0.880
2.0	3	1179	0.870
2.0	4	1179	0.865

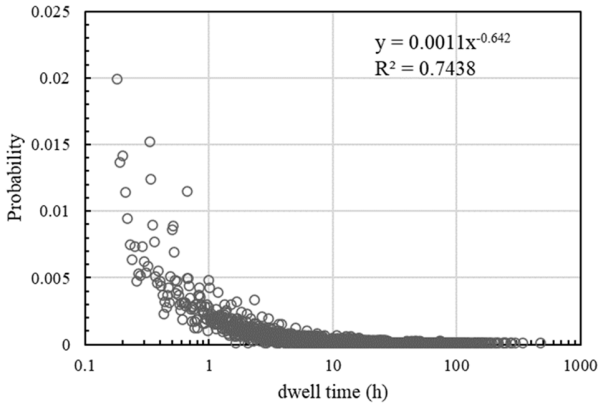


Fig. 7. The dwell time distribution.

useful metric for evaluating the quality of clustering results with different algorithm parameters. It ranges from -1 to 1, where negative values typically indicate that samples from different clusters are more similar, suggesting incorrect clustering assignments. Positive values indicate reasonable sample clustering, with higher values indicating better clustering results. Table III presents the clustering results of truck trajectories under different parameters, where eps is set to 2km and MinPtS is set to 2, resulting in better clustering performance. In comparison to previous parameter settings for individual trajectory clustering, such as Eps =25m, MinPtS =4 [35], and Eps =72m, MinPtS =545 [36], our approach extracts cluster points based on idle GPS points, resulting in sparser truck trajectories. Consequently, a larger Eps range and fewer MinPtS are used.

The centroid of the stop is determined by taking the average latitude and longitude of the cluster, while the dwell time in the stop is calculated based on the timestamps of the first and last points within the cluster. In Fig.7, the dwell time distribution is depicted, showing a power law distribution. The probability density of the dwell time follows the equation $y = 0.0011x^{-0.642}$, R^2 =0.7428. This indicates that the majority of stops are short-term stops, often occurring temporarily during the journey, while longer stops are more likely to be the final destination.

2) Truck Trip Identification: Truck stops during trips are typically short in duration, often due to factors like traffic lights or brief traffic congestion [15]. However, at the end of a trip, trucks often spend a significant amount of time loading

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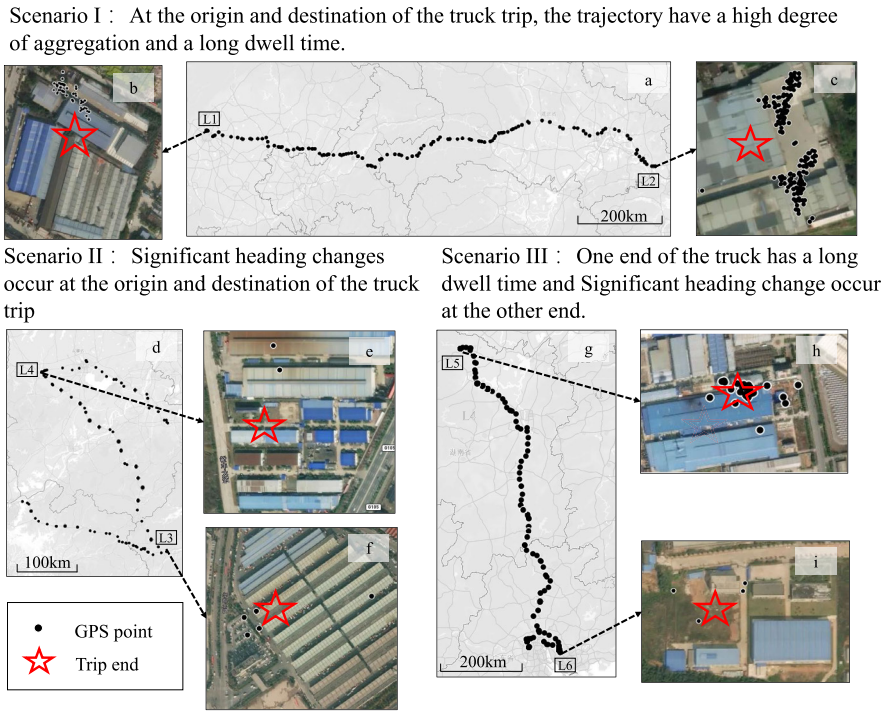


Fig. 8. Example of a truck trip identification scenario.

and unloading. Existing algorithms for identifying the end of a truck trip primarily rely on the dwell time of stops [7]. In Fig. 8, scenario I illustrates a truck that has spent a long time at both the origin location L1 and the destination location L2. Based on testing with smaller trajectory datasets containing trip information, this study selects a dwell time threshold of 55 minutes.

Furthermore, it was observed that there are still some trip ends with relatively short dwell times that cannot be effectively identified using the dwell time threshold alone. However, the change in heading provides an effective criterion for identification. In Fig. 8, scenario II shows that the truck stays for a shorter time at both the origin location L3 and the destination location L4, but there is a significant change in heading between these two points. Similarly, scenario III demonstrates a longer stay at the origin location L5 and a shorter stay at the destination location L6, but there is a notable change in heading at the destination location L6.

Due to the time-sensitive nature of truck travel, the revenue generated by individual trucks is primarily based on the distance traveled rather than the time taken [16]. When a truck does not reach its intended destination, it typically moves in a nearly straight line to expedite arrival and improve time profitability. Once the transportation task is completed, the truck changes direction to initiate the next trip. Consequently, significant changes in heading before and after a particular position during truck travel often indicate the end of a trip.

Previous studies have utilized GPS points to identify individual trip ends based on heading changes [37]. However, this approach is susceptible to the influence of road topology, such as winding mountain roads, as the heading change between GPS points can be heavily affected. To address this issue, this study calculates the heading change between stops, as illustrated in Fig. 9. By enlarging the observation scale, this method circumvents the influence of road topology, resulting in more accurate trip end identification.

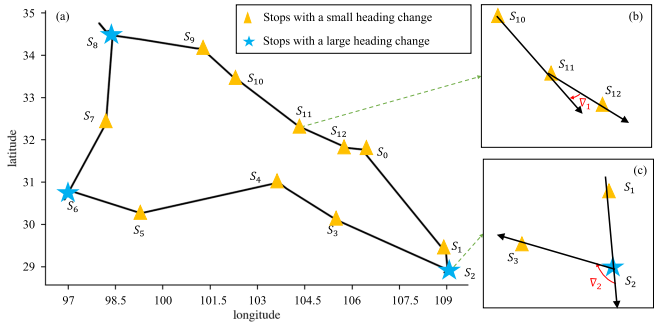


Fig. 9. Overview of the heading change of the truck stops.

To calculate the heading change, we utilize the latitude and longitude coordinates of truck stops. Firstly, we convert these coordinates into Cartesian coordinate values using formulas 1-3, where r represents the radius of the Earth. Subsequently, we construct a vector using formula 4. By considering the dot product relationship between vectors of two consecutive stops, we can determine the angle of heading change using formula 5.

$$x = r * \cos\left(lat * \frac{\pi}{180}\right) * \cos\left(lon * \frac{\pi}{180}\right) \quad (1)$$

$$y = r * \cos\left(lat * \frac{\pi}{180}\right) * \sin\left(lon * \frac{\pi}{180}\right) \quad (2)$$

$$z = r * \sin\left(lat * \frac{\pi}{180}\right) \quad (3)$$

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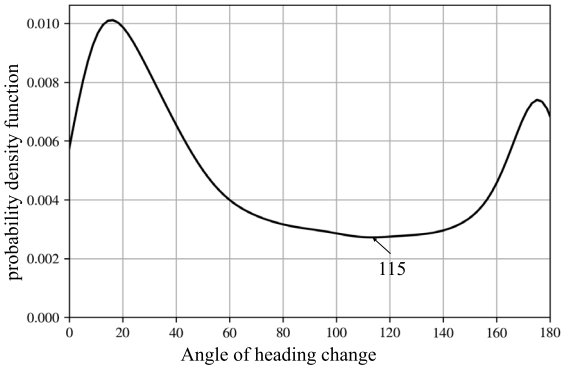


Fig. 10. The angle distribution of heading change between the continuous truck stops.

$$v_t = [x_t, y_t, z_t]\,[\vec{x_{t+1}}, y_{t+1}, z_{t+1}] \tag{4}$$

$$\nabla = acos\left(\frac{v_{t-1} \bullet v_t}{||v_{t-1}||||v_t||}\right) * 180 \tag{5}$$

Fig.10 illustrates the distribution of heading changes for truck stops. It can be seen that the angle of heading change exhibits a bi-polar bimodal distribution, indicating that when the geographical location of the truck's stop changes, it predominantly occurs either in a nearly straight line at an angle of less than 40 degrees or in a sharp turn at an angle exceeding 160 degrees. This finding aligns with our initial expectations. Based on the analysis of the heading change distribution of truck stops, this study sets the threshold for heading change as the saddle point of the probability density, which is determined to be 115 degrees. Consequently, a truck end is identified when the truck stop has a dwell time exceeding 55 minutes or a heading change exceeding 115 degrees.

There may be instances where rest activities in service areas result in a longer dwell time or turn-back stops occur during truck transportation, leading to significant heading changes (e.g. when the driver takes the wrong route). If the dwell time or heading change exceeds the set thresholds, the method may incorrectly identify these stops as trip ends. To address this issue, a effective correction method based on GIS information is employed to eliminate trip ends within road areas [36].

In this study, the road network information from the Open Street Map [38] is utilized. A 20-meter buffer is applied to the three main road types in the study area: motorway, trunk, and primary. Additionally, the Amap application programming interface (https://lbs.amap.com) is used to retrieve the locations of rest areas. If a trip end falls within these areas, it is removed from the identified trip ends. Finally, consecutive trip ends are connected to form complete truck trips. If two consecutive trip ends fall within the administrative boundaries of different cities, the trip is classified as an intercity trip.

D. Charging Station Location Model Based on Truck Trips

When the battery's range is insufficient to meet the required driving distance for intercity truck trips, recharging on the way is a common solution [26]. However, due to limited government budgets and the high cost of infrastructure construction, the location of en-route charging infrastructure is crucial. We're curious about how many facilities are needed for BET drivers to complete intercity trips without incurring excessive detour costs. Therefore, we take into account battery range and charging infrastructure budget limits, and develop an optimal location model for truck charging infrastructure. This model will provide drivers with an energy-viable path from origin to destination, while minimize driver detour mileage. For discussion convenience, the notation is presented in Table IV.

TABLE IV
NOTATION

Sets	
W	Set of origin-destination pairs, where W={o-d}
G	network G =(N, A) consisting of sets of nodes N and links A
N	Set of nodes, where N={n}
A	Set of links, where A={(i, j)}, where i, j ∈ N
O	Set of all origin nodes o, where O ⊂ N
D	Set of all destination nodes d, where D ⊂ N
variables	
z_t	Node construction indicator that indicates whether a charging station is built or not, where $z_t = 1$ if a charging station is constructed at candidate node t, and otherwise $z_t = 0$
x_{ij}^{od}	Activation indicator of link (i,j), where link (i,j) can be used to carry traffic flow of origin-destination pair o-d if $x_{ij}^{od} = 1$, and otherwise if $x_{ij}^{od} = 0$
parameters	
dis_{ij}	The length of the link (i, j)
dis^{od}	The length of origin-destination pair o-d
B	The maximum number of charging stations to be built
R	The driving range limitation of a BET after a full charge

This paper focuses on addressing the charging needs of intercity trucks on the way. Given the abundance of urban charging facilities, we assume that all electric trucks are fully charged before their departure [26]. In general, the electricity consumption of electric vehicles is assumed to be approximately proportional to the distance driven, and studies have shown that the speed of power consumption is about 1.125 kWh/km [4], based on this, we convert the power limit to a distance limit. And we use the real navigation distance to calculate the distance between two locations, the advantage of this processing is that the calculation is more realistic, due to the distance between some network nodes being much greater than the straight-line distance with poor road connectivity, moreover, simplifying the model processing process. The model is as follows:

$$min \sum\nolimits_{o-d \in W} \left(\sum\nolimits_{(i,j) \in A} dis_{ij} * x_{ij}^{od} - dis^{od}\right) \tag{6}$$

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subject to Constraint 7-12.

$$\sum_{\{j:(i,j)\in A\}} x_{ij}^{od} - \sum_{\{j:(i,j)\in A\}} x_{ji}^{od} = \begin{cases} 1, i \in O, \\ 0, i \in N\backslash\{O\cup D\}, \\ -1, i \in D, \end{cases} \quad \forall o-d \in W \tag{7}$$

$$\sum_{t\in N\backslash\{O\cup D\}} z_t \le B \tag{8}$$

$$dis_{ij} * x_{ij}^{od} \le R, \forall (i,j) \in A, \forall o-d \in W \tag{9}$$

$$x_{ij}^{od} \le z_i * z_j, \forall (i,j) \in A, \forall o-d \in W \tag{10}$$

$$x_{ij}^{od} \in \{0,1\}, \forall (i,j) \in A, \forall o-d \in W \tag{11}$$

$$z_t \in \begin{cases} \{0,1\}, \forall t \in N\backslash\{O\cup D\} \\ \{1\}, \forall t \in \{O\cup D\} \end{cases} \tag{12}$$

The objective function (6) aims to minimize the detour distance for all trips. Constraint 7 is a flow balancing constraint. Constraint 8 imposes a budget constraint on the maximum number of charging infrastructure sites that can be constructed, assuming equal construction costs for all candidate sites. Constraint 9 limits the distance between two consecutive charging stations used to carry the traffic flow of an origin-destination pair o-d to be within the battery range limitation of the BET. Constraint 10 restricts nodes i and j to build charging stations before the link (i, j) can carry the traffic flow. Lastly, Constraints 11 and 12 are binary constraints that enforce the decision variables to be either 0 or 1.

IV. Results And Discussion

A. Data Validation

1) Method Accuracy: Since the real trip origins and destinations of trucks are often difficult to obtain in practice [33], the current predominant validation method primarily relies on artificial visual verification using satellite maps [7], [13], [15], which is a time-consuming and inevitably subjective process.

The trip dataset obtained from the OFEX platform provides real trip information that can serve as ground truth for verifying and comparing the accuracy and reliability of the truck trip identification algorithm. For smaller trajectory datasets with trip records, we constructed a confusion matrix to evaluate the accuracy of trip end identification. The confusion matrix, as illustrated in Table V, is defined as follows:

TP (True Positive): The number of true trip ends correctly identified as trip ends.

FN (False Negative): The number of true trip ends incorrectly identified as false trip ends.

FP (False Positive): The number of false trip ends incorrectly identified as trip ends.

TN (True Negative): The number of false trip ends correctly identified as false trip ends.

Using the recall metric to measure how many true trip ends are correctly identified:

$$Recall(R) = \frac{TP}{TP + FN} \tag{13}$$

Using the Precision metric to measure how many of the samples predicted as the end of the trip are true trip ends:

$$Precision(P) = \frac{TP}{TP + FP} \tag{14}$$

TABLE V
The Confusion Matrix for the Trip End

	Predicted as Positive	Predicted as Negative
Labeled as Positive	True Positive (TP)	False Negative (FN)
Labeled as Negative	False Positive (FP)	True Negative (TN)

TP (True Positive): The number of true trip ends correctly identified as trip ends.

FN (False Negative): The number of true trip ends incorrectly identified as false trip ends.

FP (False Positive): The number of false trip ends incorrectly identified as trip ends.

TN (True Negative): The number of false trip ends correctly identified as false trip ends.

F1 is a performance metric that considers both precision and recall.

$$F1Score(F1) = \frac{2PR}{P + R} \tag{15}$$

Fig.11 presents the trip-end identification accuracy under different parameters. Specifically, when considering trips with a dwell time longer than 55 minutes, the algorithm achieves an F1 value of 0.8859. When incorporating both a dwell time longer than 55 minutes and a heading change greater than 115 degrees as criteria, the truck trip identification algorithm exhibits the best performance, resulting in an improved F1 value of 0.8960. This suggests that considering both parameters leads to even better accuracy in identifying truck trips. Furthermore, the inclusion of the heading change as a parameter leads to a significant increase in the recall (R) by approximately 4%. This demonstrates that the heading change at truck stops plays a crucial role in accurately identifying truck trip ends.

2) Comparison With Benchmark Methods: Since this paper uses the driver's mobile GPS data, there is a certain difference from the previous vehicle's GPS data to infer the truck trip ends (e.g. the lack of truck engine information of driver's mobile GPS data), therefore, this paper mainly selects algorithms of common datasets as the benchmark. Table VI shows the accuracy of the truck trip identification method proposed in this paper compared to previous studies. The previous methods, [5], [6], [7] of setting the speed threshold to 0 are less effective. Ideally, the truck trajectory points with a speed of 0 should be set as the truck stop point, but since this paper uses the truck driver's mobile phone trajectory data, the instantaneous speed of the truck is lacking, and the driver's mobile phone trajectory may still be in a low-speed state when the truck stopped, so setting the speed threshold to 0 is not applicable to the dataset in this paper.

Among these benchmarks, experiment 6 [11], [19] chose a speed threshold of less than 8km/h (i.e. 5mph) and a duration of more than 30 minutes was better, with an F1 score of 0.7811. In comparison to the top-performing benchmark (experiment 6), the F1 value of our algorithm (experiment 1) increased by 11.49%. This improvement can be attributed to

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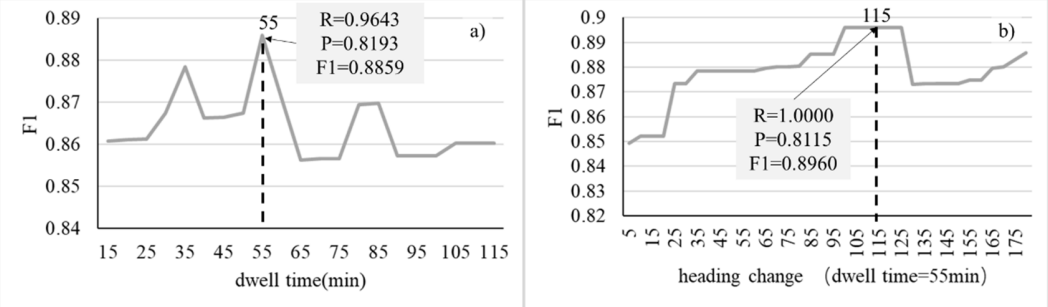


Fig. 11. Trip end identification accuracy under different parameters.

TABLE VI
TRUCK TRIP IDENTIFICATION RESULTS

Experiment	Stop identification		Trip end identification		R	P	F1	Studies
	Speed(km/h)	DBSCAN	Dwell time(min)	Heading change				
1	**15**	Eps =2km MinPtS =2	**55**	**115**	**1.0000**	**0.8115**	**0.8960**	This paper
2	0	-	3	-	0.2048	0.8185	0.3276	[5]
3	0	-	26	-	0.6586	0.7567	0.7042	[7]
4	0	-	120	-	0.0458	0.9167	0.0873	[6]
5	10	-	5	-	1.0000	0.4822	0.6507	[9]
6	8	-	30	-	1.0000	0.6408	0.7811	[11], [19]
7	5	-	10	-	1.0000	0.4776	0.6464	[8]

the effective elimination of invalid stays through the spatial clustering of the idle trajectory segment, resulting in a significant enhancement in the P value. The comparison with the benchmark clearly indicates that the truck trip identification algorithm proposed in this paper achieves a higher level of accuracy.

B. Truck Travel Characteristics and Charging Needs in Intercity Truck Trips

The dominant factors that significantly impact truck drivers' range anxiety and the adoption of BET in long-haul intercity transportation scenarios are widely acknowledged to be the battery-range limitation and the availability of charging opportunities within the transportation network. The battery range limitation is primarily determined by the energy density and size of the battery. The availability of charging opportunities is contingent upon the investment policies for charging facilities. A more efficient layout of the charging infrastructure can result in an increase in the number of charging opportunities.

To accelerate the development of truck electrification, it is necessary not only to consider technological improvements but also to understand the driving needs of trucks, which can help to develop more effective technical strategies. In this context, we analyze the compatibility of truck travel characteristics and electric vehicle technologies requirements, and what level of BET technologies needs to be raised.

1) Battery-Range Requirements for Electric Trucks in Intercity Trips: We first calculate the intercity truck trip distance dis_{od}. Fig.12 depicts the distribution of intercity truck trip

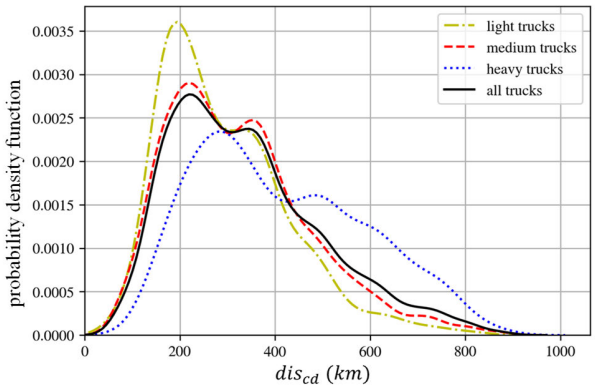


Fig. 12. The distance distribution of intercity truck trips.

distances. The solid black line represents the overall distribution, revealing two prominent peaks at approximately 220km and 350km, respectively. The other three dotted lines indicate the distance distribution of light, medium, and heavy trucks. The trip distance range of light trucks is concentrated at about 200km, medium trucks are concentrated at 220km and 350km, and heavy trucks are concentrated at 300km and 500km. Previous research suggests that the current battery capacity of electric vehicles is approximately 282kWh, with a mileage consumption of 1.125kWh/km [4]. Consequently, the battery range limitations are around 250km. This implies that the current battery range limitations are insufficient to meet the transportation distance requirements of intercity trips for light, medium, and heavy trucks.

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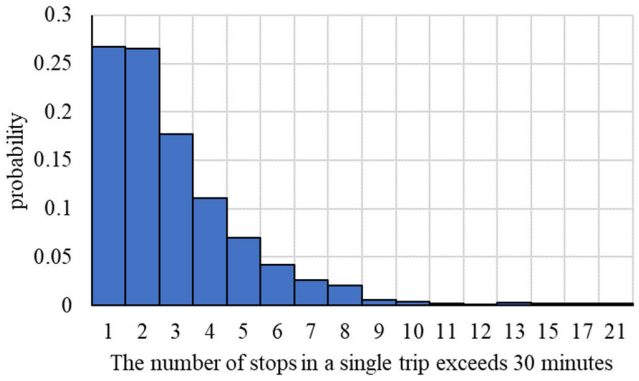


Fig. 13. The number of truck stops over 30 minutes distribution in intercity trips.

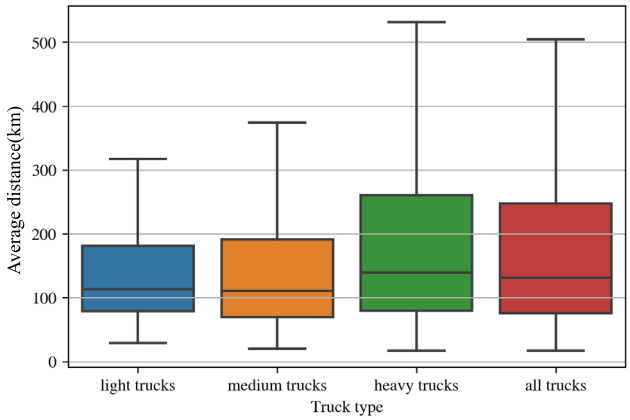


Fig. 14. The average driving distance between stops with charging opportunities.

However, truck travel often involves multiple stops, including rest breaks and mandated government stops. These stops present an opportunity for charging along the way without requiring additional effort. It is reasonable to assume that drivers would not charge during short stops due to the charging uncertainty (such as queuing, or charging takes a long time). Existing research commonly suggests that vehicle charging can occur during stops lasting more than 30 minutes [39]. Fig.13 shows the frequency of stops lasting over 30 minutes during intercity truck trips, with the majority of stops concentrated between 1 to 3 times.

The average driving distance between stops with charging opportunities, i.e. those lasting over 30 minutes as depicted in Fig 14, provides an approximate measure of the distance traveled by the truck before taking a break that could potentially allow for recharging. It is evident that 75% of the average driving distance between charging stops for light trucks and medium trucks is less than 200km, while heavy trucks have a longer average driving distance of about 250km. This observation highlights the fact that by considering the distance traveled between two consecutive charging stops instead of the entire trip, the minimum battery range requirement for electric trucks can be significantly reduced.

The driving distance between consecutive stops lasting over 30 minutes, referred to as the continuous driving distance

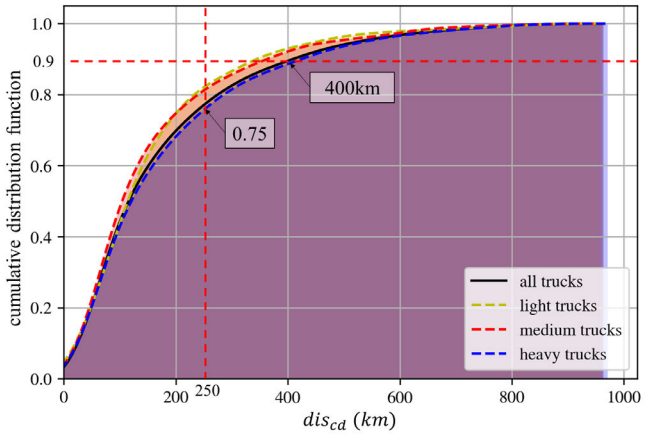


Fig. 15. The cumulative probability distribution of the dis_{cd} of intercity trips.

dis_{cd}. Fig.15 presents the cumulative probability distribution of the dis_{cd} of intercity truck trips. It is observed that there is a 75% probability of the continuous driving distance being less than 250 km, and over 90% of the continuous driving distance being less than 400 km. These findings indicate that the current battery range limitations are able to accommodate 75% of the continuous driving distance. As electric vehicle technology advances, reaching a battery range of 400 km would cover nearly 90% of the continuous driving needs for intercity trucks. When comparing different types of trucks, similar driving characteristics are observed, which can be attributed to the similarity in living habits and policy impacts within truck groups.

Moreover, our study specifically examines the continuous driving behavior during the portion of the intercity trips where the driving distance dis_{od} exceeds 250km. This particular segment is of utmost importance when considering battery range design. We use some common distribution functions to fit the continuous driving distance dis_{cd} of these trips with dis_{od} greater than 250km. Choose the distribution function with the smallest residual squares $RSS = \sum_{k=1}^{n} (y_k - f(x_k))^2$, where, y_k is the ith value of the variable to be predicted, x_k is the ith value of the explanatory variable, and f(xi) is the predicted value of yi. The results show that the continuous driving distance dis_{cd} of these intercity truck trips follows the BETA distribution (a=1.111, b=25.0462, loc=20.0571, scale =5614.21), as shown in Fig.16. This analysis can provide valuable insights for informing more precise battery range design. For instance, it can assist decision-makers in developing tailored battery standards for different electrified truck replacement scenarios, thereby catering to the specific needs of a more targeted truck market segment.

To further validate the beta distribution, we employed the Kolmogorov-Smirnov test (K-S test) to examine the parameters of beta (a=1.111, b=25.0462, loc=20.0571, scale=5614.21), which is a useful nonparametric hypothesis test. This test is commonly used to determine whether a set of samples originates from a specific probability distribution (one-sample K-S test) or to compare the similarity of the distributions of two sample sets (two-sample K-S test). The

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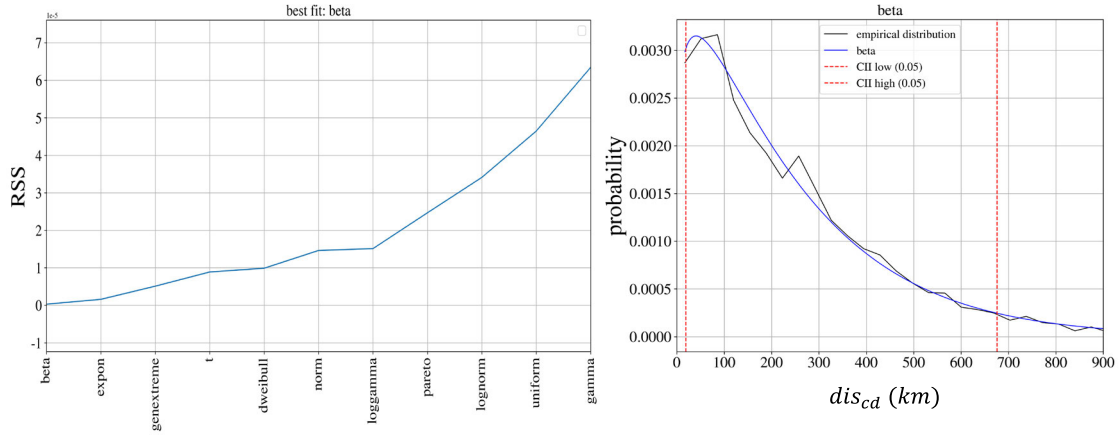


Fig. 16. Fitting of the continuous driving distance distribution for intercity truck trips with dis_{od} longer than 250km: (left) RSS for different probability distributions, (right) probability distribution for beta distribution.

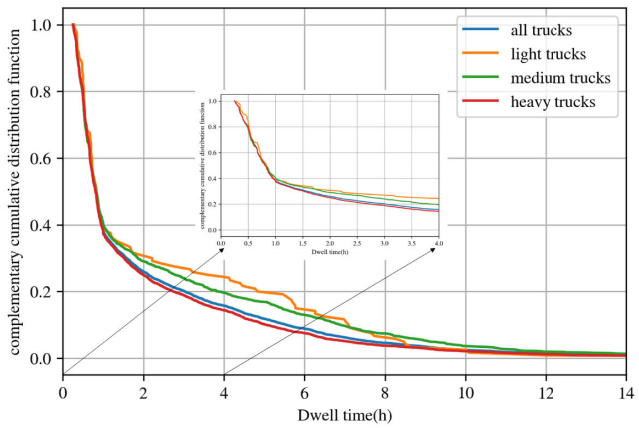


Fig. 17. The Complementary Cumulative Distribution Function of intercity truck dwell time.

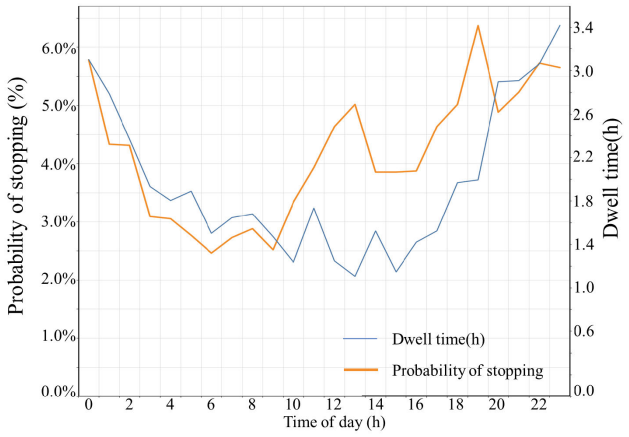


Fig. 18. The distribution of periods intercity trucks stopped.

results of the K-S test indicated that the p=0.011, which is lower than the significance level of 0.05. This suggests that the continuous driving distance distribution for intercity truck trips exceeding 250km adheres to the beta distribution.

2) Charging Power Requirements of Charging Facilities for Intercity Truck Trips: Currently, the charging rate of charging

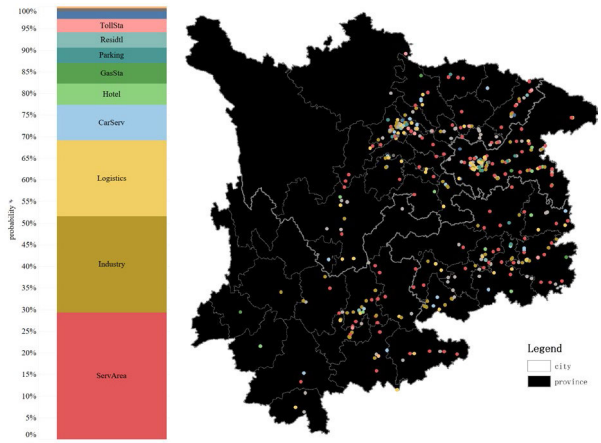


Fig. 19. The distribution of truck stopping locations.

stations is continuously advancing. In China, the predominant charging power is approximately 120 kW. Considering a truck battery capacity of 282 kWh, theoretically, it would take over 2 hours for a truck to be fully charged using a 120 kW facility. If the disparity between the time needed for a full charge and the typical duration of truck stops is substantial, this would result in additional time expenses for truck drivers. Consequently, this impedes time-sensitive truck drivers from adopting BET as a viable alternative.

Understanding the distribution of truck dwell time can inform the design of charging power at charging stations that offer on-the-way charging services for intercity truck trips. Fig.17 illustrates the Complementary Cumulative Distribution Function of truck stop dwell time. The blue line represents the dwell time distribution of all trucks, while the other three lines represent the dwell time distribution of different truck types. It is evident that approximately 80% of the probability of truck dwell time exceeds 30 minutes, whereas only 40% of the probability of truck dwell time exceeds 1 hour. If a 282 kWh battery can be fully charged within half an hour, the charging power would need to reach 564 kW, which is consistent across different truck types.

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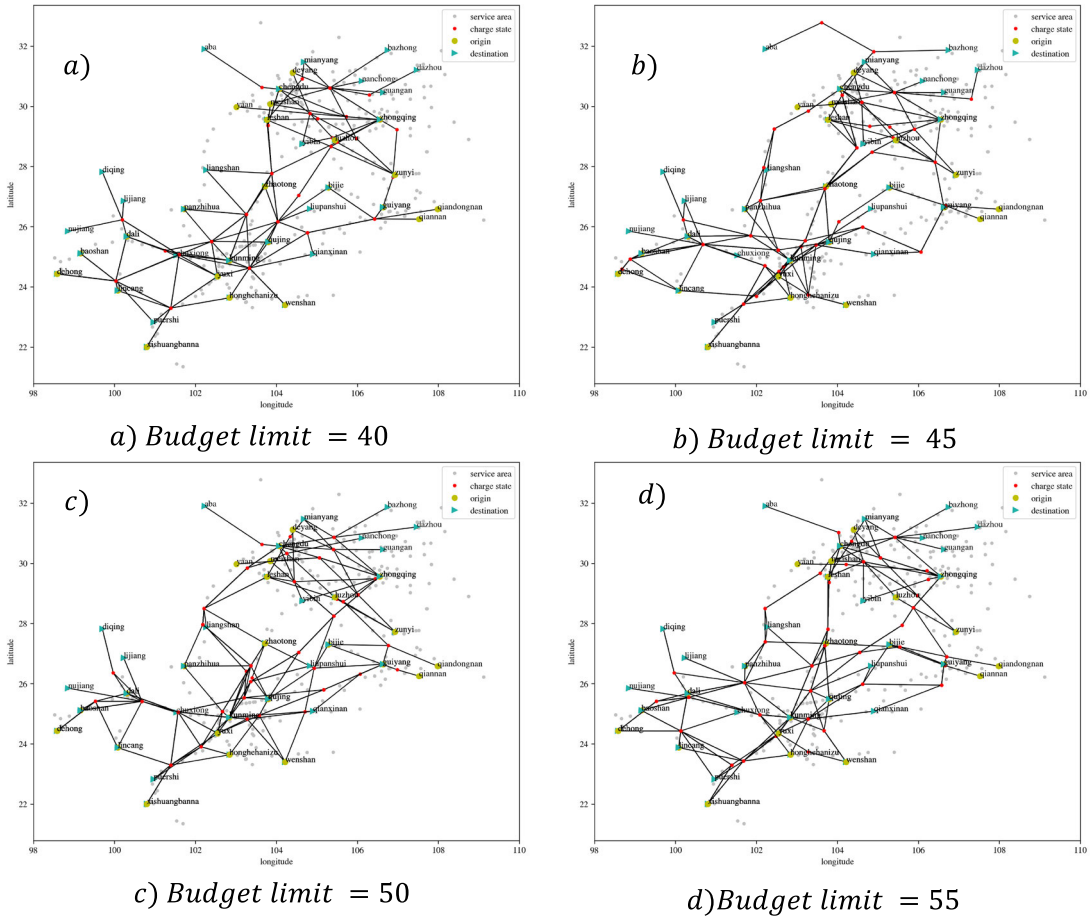


Fig. 20. The optimal charging station location and the route choices of intercity trucks under different budget limits in southwest China.

Another concern is the potential impact of sudden load on the main electrical grid, which is closely related to the probability of truck groups making stops. To analyze this, we collected data on the start time (measured in hours) and the median dwell time of truck stops. As shown in Fig.18, the start time of intercity trucks exhibits three distinct peak periods: 11:00-13:00, 18:00-19:00, and 21:00-1:00. These peaks align with typical driver living habits, with 11:00-13:00 and 18:00-19:00 being lunch and dinner times for truck drivers, respectively, and the median dwell time during these periods is approximately 1-2 hours. The period from 21:00-1:00 represents the truck drivers' sleep time, with a median dwell time of around 3 hours. During these peak stop hours, the potential charging behavior of BET drivers may impose a significant load on the main electrical grid, resulting in insufficient charging power.

3) The Layout of Charging Facilities for Intercity Truck Trips: The layout of charging infrastructure plays a crucial role in the development of electric trucks, as it represents a significant portion of BET investment. A more efficient charging infrastructure layout can lead to more charging opportunities to relieve or even tame range anxiety for BET drivers. We believe that designing the infrastructure layout based on drivers' parking characteristics can enhance acceptance among BET truck drivers. To achieve this, we first examine the distribution of truck-stopping locations to identify frequent stopping points. Fig.19 illustrates the distribution of truck stopping locations where intercity trucks spend more than 30 minutes. Among these stops, 30% are at service areas, followed by factories and logistics centers. It is evident that service areas, compared to other hotspots, offer more rest services for truck drivers and are thus more suitable for charging infrastructure placement.

We then proceed with the selection of charging facility locations based on real trip data using the model proposed in section III-C. From the extracted intercity truck trips in southwest China as discussed in section III-D, we choose 7,000 long-distance intercity truck trips with dis_{od} longer than 250km, resulting in a total of 180 different city OD pairs. These service areas where intercity trucks have stopped for more than 30 minutes are considered as potential charging station locations, totaling 302 locations.

Given the large size of the alternative charging station location set, the charging station location model proposed in section III-D has high calculation complexity. Therefore, we employ the Genetic Algorithm to determine the optimal set of charging station locations. Fig.20 illustrates the route choices of intercity truck trips under various budget limits. The gray dot represents the geographical location of the alternative charging station, the green dot represents the departure city of the intercity truck trip, the blue triangle represents the destination city of the intercity truck trip, and the red dot represents the selected optimal set of charging station locations.

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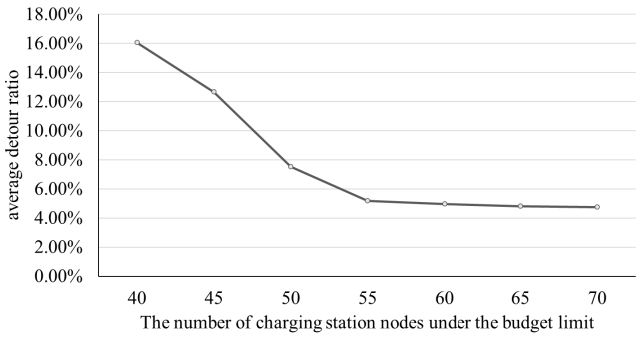


Fig. 21. Average detour ratio of intercity truck trips under different budget constraints in southwest China.

By implementing these charging stations, we can provide an energy-viable route for all these long-distance intercity truck trips.

Additionally, we investigate the number of facilities required to offer BET drivers an energy-viable route for intercity trips, while minimizing detour costs. We consider different limits for the number of charging stations, ranging from 40 to 70 in increments of 5. To assess the impact of charging behavior on detour costs, we introduce a detour indicator, denoted as average_detour_ratio, which calculates the average detour ratio of all trips. This indicator takes into account both the route distance and the actual driving distance of each trip.

$$\text{average_detour_ratio} = \sum_{i}^{n} \frac{dis^{od}_{route} - dis^{od}_{actual}}{dis^{od}_{actual}} \tag{16}$$

Fig.21 illustrates the average detour ratio of intercity truck trips in southwest China under different budget constraints. When the number of charging stations is set to 40, the average detour ratio is approximately 16%. However, as the number of charging stations increases to 55, the detour ratio decreases to around 5%. This indicates that even though there are over 300 alternative charging station locations available for selection within the study area, choosing a subset of 55 charging stations can enable BET to complete these 7,000 intercity trips with a low detour ratio. These experimental results demonstrate that selecting a small subset of alternative charging facility locations (55/302) can provide BET drivers with an energy-viable path to complete intercity trips without incurring excessive detour costs.

V. Conclusion

A. Electric Trucks Are Struggling to Complete Intercity Trips

By analyzing the travel characteristics of nearly 900 truck drivers over two months, we identified intercity truck trips and stops. The identification of truck stops was accomplished by examining parameters such as speed and trajectory density. Additionally, the identification of truck trips was achieved by utilizing truck heading changes as a distinguishing feature, supplemented by a dwell time threshold. The validation results showed that our method improved both recall and accuracy. Based on our findings, we observed the following key points:

(1) The trip distances for light trucks tend to concentrate around 200km, while heavy trucks at 300km and 500km. However, the current limitations of battery range present challenges for electric trucks to meet the distance requirements of intercity trips.

(2) Considering the current charging power, it takes over 2 hours for a truck to achieve a full charge. However, only 40% of truck stops en-route last longer than 1 hour. This implies that truck drivers would need to allocate additional time for charging, which could be inconvenient for time-sensitive drivers and hinder the practical use of electric vehicles as a viable substitute.

(3) Insufficient en-route charging stations for intercity trucks not only impact the convenience of driving electric trucks but also reduce the willingness of potential consumers to purchase them.

B. Technical Development Requirements for Electric Trucks for Intercity Trips

Our analysis of intercity truck travel behavior has identified several key technical development requirements for electric trucks, including the following:

(1) Battery range: Our findings indicate that a battery range of 250km would be sufficient to cover 75% of the continuous driving needs of trucks, while a battery range of 400km would cover nearly 90% of the continuous driving needs of intercity trucks. The distribution of continuous driving ranges for long-distance intercity truck trips follows a beta distribution.

(2) Charging power: We have observed that approximately 80% of truck dwell time falls within the 30-minute range. If a conventional battery with a capacity of 282kWh can be fully charged within this timeframe, it would require a charging power of over 564kW. Additionally, there may be high load demands on the intercity power grid during specific time periods, such as 11:00-13:00, 18:00-19:00, and 21:00-1:00. This could result in potential peak charging hours, leading to insufficient charging power availability.

(3) Charging facility layout: Our experimental results demonstrate that by selecting a small subset of alternative charging facility locations (55/302) based on travel behavior, the detour ratio for truck charging can be reduced to just 5%, offering BET drivers with an energy-viable path to complete intercity trips without excessive detour costs. This indicates that the travel behavior-based charging station layout optimization model can effectively reduce travel costs for truck drivers.

C. Policy Implications

(1) For policymakers advocating for truck electrification

Firstly, to promote the widespread adoption of electric trucks for intercity trips, it is crucial to address the current limitations in battery range and charging power. The government needs to invest in research and development to improve the technology and provide incentives for companies to adopt electric trucks. Additionally, power deployment during peak parking periods may be necessary to ensure stable charging services.

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Secondly, it is recommended to build charging facilities according to the travel behavior of trucks to increase truck acceptance and reduce detours. In cases where detours cannot be avoided, the government may need to offer subsidies to offset the costs incurred by time-sensitive truck drivers. These policy recommendations can facilitate the transition to electric trucks for intercity trips and support the growth of the electric vehicles industry.

(2) For electric truck manufacturing-related enterprises

Firstly, we have analyzed the range requirements in truck travel, but in practice, we need to consider the fear and uncertainty experienced by drivers who worry about running out of power before reaching their destination. Electric truck manufacturers need to provide an additional range to alleviate driver's range anxiety. By doing so, help them feel more confident in using electric trucks for their transportation needs.

Secondly, we conducted an analysis of the peak periods for truck stopping probability on the road network. During these peak periods, it is imperative to fully consider the usability of charging stations. The platform reservation mechanism can be utilized to allocate charging stations in advance and enhance the efficiency of truck drivers' charging experience. This advanced system offers numerous benefits beyond just efficient allocation. It enables better coordination between truck drivers and charging station operators by providing real-time information about availability and expected waiting times.

Thirdly, it is recommended to install more charging stations at strategic locations along popular road freight routes to cater to the increasing demand for charging services. This will allow larger volumes of vehicles to be accommodated simultaneously, minimizing wait times. Furthermore, strategically placed charging stations will serve as essential hubs for drivers during their breaks or layovers. Offering amenities such as comfortable waiting areas with Wi-Fi connectivity and refreshment options would enhance the driver experience and promote adoption of electric trucks within the industry.

D. Limitations and Future Research

There are some limitations to the research that should be acknowledged. Firstly, this study aims to evaluate the technical requirements of electrified vehicles based on individual truck travel demand, while neglecting the interdependence among these technical specifications. Secondly, in the long-haul intercity truck transportation scenario, electric trucks in various regions of the world are still in the early stage of development, and there is a general lack of actual travel data for electric trucks. The empirical data used in the study is derived from gasoline-enabled vehicles, which may impose limitations on the generalizability of the findings. Thirdly, similar to previous studies [4], [26], [40], we made the assumption that the electricity consumption of electric trucks is approximately proportional to the distance traveled (i.e., 1.125 kWh/km), which has been proved by previous studies of measuring electricity consumption [41], [42]. Indeed, there exist various other factors that impact the rate of energy consumption, such as driving behaviors, vehicle characteristics, vehicle operations, environmental conditions, and system-level technologies [4]. However, for the purpose of simplifying the research model, these factors were not considered in our study, this simplification may limit the applicability of the results in practical scenarios that require a more nuanced understanding of energy consumption.

Besides these limitations, this study fills a significant knowledge gap by examining the travel requirements of intercity electric trucks. It provides a comparative analysis of the demand for intercity truck travel and the current state of electric vehicle technology, thus offering novel insights into the technical prerequisites of electric trucks based on real-world data. The study quantified the necessary enhancements in battery range capabilities, charging power, and charging station layout to facilitate the successful implementation of intercity freight transportation services using electric trucks. Additionally, the research offers valuable policy recommendations aimed at promoting the adoption of electric trucks in this context.

Based on our research findings, future studies could explore the implementation of collaborative modeling among key technologies in electric trucks to optimize the power assurance capability for long-haul intercity trips. For example, increasing the battery range of electric trucks may reduce the need for high charging speeds at charging stations. A more comprehensive evaluation that considers the interplay between battery range, charging power, and charging facility layout could lead to more optimal solutions. In addition, upcoming investigations could implement electricity consumption models accounting for some easy-to-collect vehicular parameters, such as travel speed and gross vehicle mass. Specifically, while travel speed could be derived directly from GPS trajectories, Weigh-In-Motion databanks [43] could be mined to associate each vehicle with its Gross Vehicular Mass (GVM). Additionally, statistical models that can infer the GVM from other basic vehicular characteristics [44] could be considered in the analysis.

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